

Disease Project Overview

Goal

The project's aim is for IB students to collaborate in small teams to answer the following question regarding a specific disease:

How can changes in gene/protein sequence and structure explain the molecular basis of disease? What can gene/protein comparison tell us about the affected biological function?

The project will require students to integrate key theoretical and practical sessions for their direct application to the disease topic assigned to their team.

Timelines and Format

The project will be formally introduced on September 17th 2026. Students will have until Monday October 28th at 18:00h sharp, local time to develop the project (content contribution/commits will be discarded beyond this moment).

The project will be developed entirely in GitHub through collaborative, versioned contributions. The expected format is detailed in the Disease Project Formal Requirements file and the BCM Template Bioinformatics file.

Students will be required to upload a Gantt chart deliverable by October 9th 18:00 local time at the latest as an intermediary delivery. Students will also be required to maintain a live Project Status document throughout the entire project development process. This Project Status document will be discussed on October 22nd, but is expected to continue live until the end of the project.

Content Integration

Students are expected to leverage the knowledge learned from at least the following sessions:

- Project Management and GitHub (theory and practice)
- Paper structure and citation (theory) and data sources (practice)
- Structural Biology and Genomics (theory) and alignments (practice)

Scope Expectations

The following topics should be considered during project development. Some topics may not be applicable to some teams. Excluding a topic from the project's scope should be justified from a clear biological or practical perspective.

- What is the cause of the disease from the gene, protein, structure and function perspectives?
- What are the differences at gene, protein, structure and function levels between a reference/healthy state and the disease-associated state?
- From a biological perspective, how are the differences between reference/healthy state and disease-state impacting the patient's function?
- What are the differences of the gene, protein, structure and function between humans and at least 2 other relevant species? What biological implications does this have?

Remember, with all the knowledge you gathered answering these questions (or the ones of your choice) you must be able to give an answer to our main goal:

How can changes in gene/protein sequence and structure explain the molecular basis of disease, and what can comparison tell us about the affected biological function?